

Mole Problems #2

Show full solutions to each of the following questions (use lined paper). Remember to put your answers in the correct number of significant figures.

1. If 4.00 moles of an element has a mass of 128 g, what would be the molar mass?
2. How many moles of oxygen (O_2) are present in 64 g of oxygen?
3. What is the mass of 3.00×10^{12} atoms of aluminum?
4. How many atoms of copper are there in 25.4g of copper?
5. What is the mass of 0.50 mol of water?
6. How many atoms are there in 58 g of SO_2 ?
7. How many moles of sodium are there in 0.27g of sodium?
8. How many molecules are there in 4.0 moles of uranium hexafluoride? (UF_6)
9. Calculate the mass of 0.269 mol of ammonium chloride? NH_4Cl
10. Calculate the number of molecules of ammonia (NH_3) present in 0.27 g of ammonia?
11. What would be the mass of 3.01×10^{23} molecules of oxygen gas?
12. Calculate the number of atoms present in 60.0 g of NaOH.
13. Calculate the number of moles present in 347.0 g of ammonium nitrate (NH_4NO_3).
14. What is the mass of 1 molecule of water?
15. How many molecules of water would there be in 1.0g of water?
16. What would be the mass of 3.01×10^{24} molecules of oxygen gas?
17. How many atoms would there be in 2.45 mol of sodium sulphate?
18. What would be the mass of 0.16 mol of helium gas (He)?
19. Calculate the number of molecules present in 109.5g of hydrogen chloride gas.

Answers

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| 1. 32.0 g/mol | 2. 2.0 mol | 3. 1.35×10^{-10} g | 4. 2.40×10^{23} atoms | 5. 9.0 g |
| 6. 1.6×10^{24} atoms | 7. 0.012 mol | 8. 2.4×10^{24} molecules | 9. 14.4 g | 10. 9.6×10^{21} |
| 11. 16.0 g | 12. 2.71×10^{24} atoms | 13. 4.34 mol | 14. 3×10^{-23} g | 15. 3.3×10^{22} molecules |
| 16. 160 g | 17. 1.03×10^{25} atoms | 18. 0.64 g | 19. 1.81×10^{24} molecules | |